

seth_dice_1.2

2025-06-08

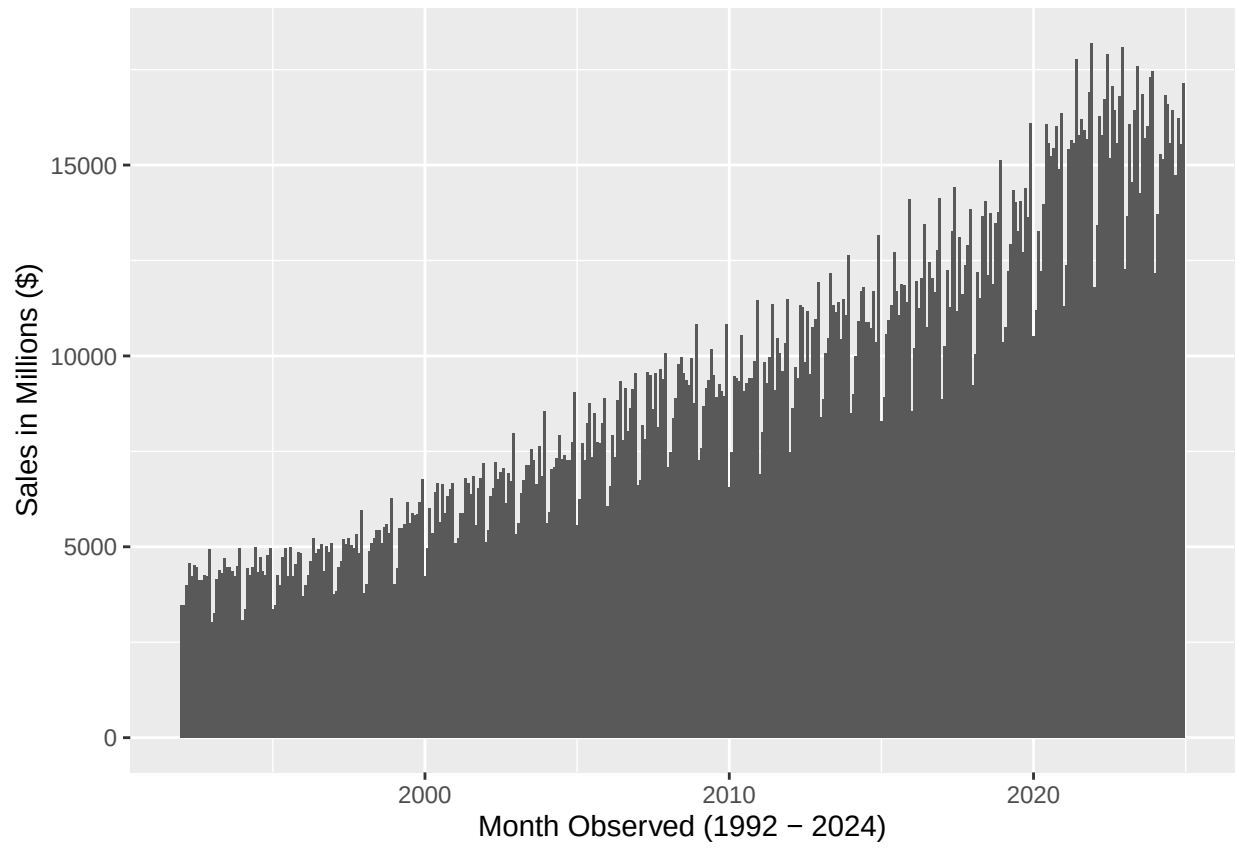
Write a summary of your data and identify at least two questions to explore visually with your data.

The data is from Federal Reserve Economic Data for total month sales in millions of dollars for alcohol of merchant wholesalers in the United States. The alcohol includes beer, wine, and distilled beverages. The "observation_date" represents the month of the alcohol sales and "millions_\$" represents the total dollar sales in millions. The total timeframe of data is from 1992-2025, but due to 2025 only having 3 months this data has been removed. The timeframe of alcohol sales is from 1992-2024.

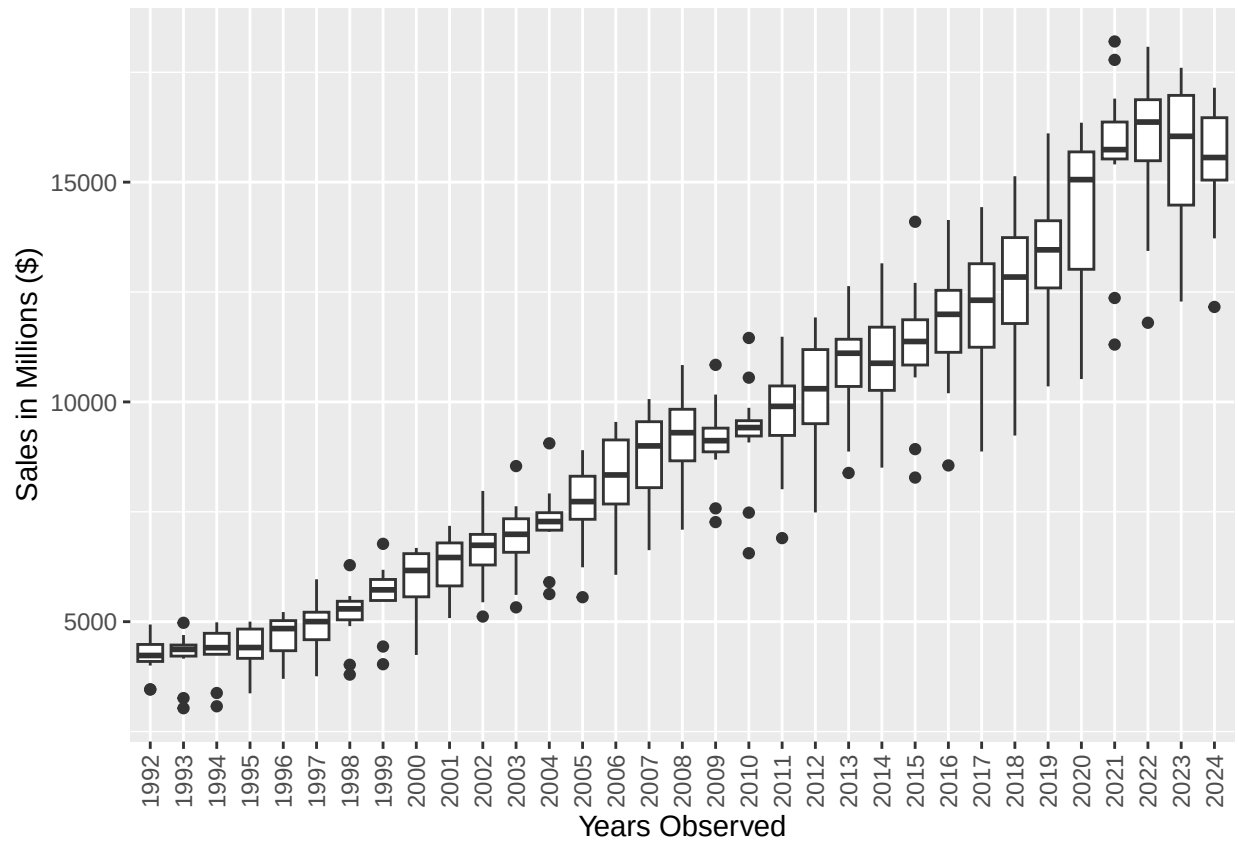
The questions I am wanting to explore with the sales data of alcohol is:

- 1) Has there been an increase in alcohol sales amongst the U.S. population?
- 2) Which months account for the highest alcohol sales amongst the U.S. population?

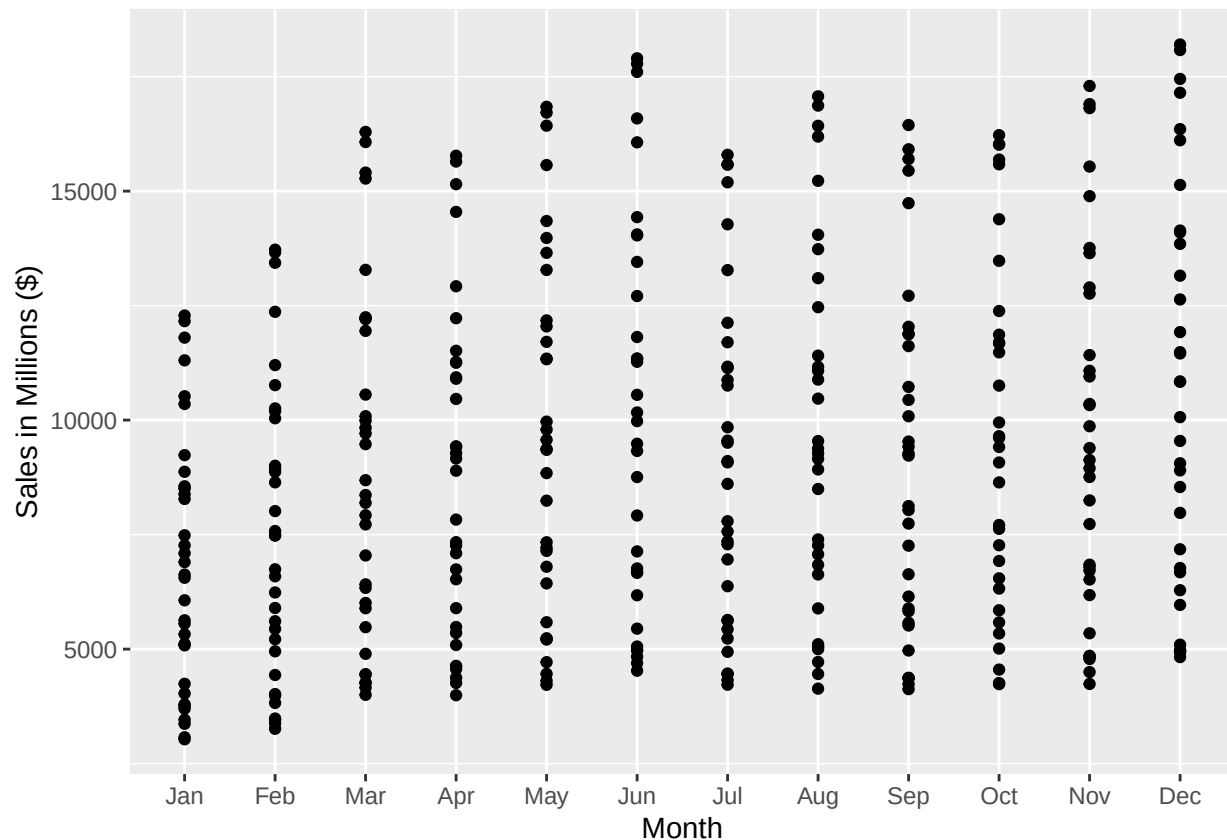
Create a histogram or bar graph from your data.



Create a boxplot from your data.



Create a bivariate plot from your data.



Summarize your results and make a conclusion. Explain how you arrived at this conclusion and how your visualizations support your conclusion.

Using the histogram and boxplot of data from 1992-2024 we can see an increase in the total alcohol sales over the years; however, from 2020 it seems to have peaked (possibly Covid) and then started to drop after. I'm curious to see if this trend continues and what may be causing the decline in alcohol sales. It could be either U.S. is starting to move away from alcohol purchases as people look to decrease their alcohol consumption (studies showing alcohol consumption is bad for you and is leading to increase number of cancers among younger population), alcohol no longer being as affordable as it once was, or different reasons.

Before reviewing the data I would have guessed the top 3 months representing the highest alcohol sales would be July (4th of July), November (Thanksgiving) and December (Christmas/New Years week after). However, reviewing the bivariate plot of total sales in millions of dollars by month the top 3 months in number of sales in millions of dollars would be December, June, and November. I'm not surprised about December and November, but I am surprised to see June. I'm guessing that June is when people are loading up on alcohol to prepare for the 4th of July. Another surprise is January and February being so low compared to the other months, but this could be due to the holiday runs of November and December with people wanting to save money... and get over their hangovers.